THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 6: **How Forces Affect Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A force of 24 N acts on a body and produces an acceleration of 4 m/s squared. The mass is: 1

(a) 6 kg **(b)** 20 kg **(c)** 0.17 kg **(d)** 96 kg

2. A person weighs 600 N on the Earth. On a planet with g one-third of the Earth's, the weight is: 1

(a) 600 N **(b)** 300 N **(c)** 1800 N **(d)** 200 N

3. A block of mass 4 kg rests on a rough table. The least force of 12 N makes it just move. Static friction is: 1

(a) 0 **(b)** 4 N **(c)** 12 N **(d)** 40 N

4. A body of mass 3 kg moving at 4 m/s is stopped in 2 s. The average force is: 1

(a) 12 N **(b)** 6 N **(c)** 3 N **(d)** 24 N

5. A person weighs 600 N on the Earth (g = 10). On a planet where g is one-third of that on Earth, the mass of the person is: 1

(a) 600 kg **(b)** 60 kg **(c)** 20 kg **(d)** 180 kg

6. A block on a rough floor can resist a push of up to 12 N by static friction. If it is pushed by a force of 8 N and does not move, the friction force is: 1

(a) 4 N **(b)** 8 N **(c)** 12 N **(d)** 0

7. A 0.5 kg ball moving at 20 m/s is stopped in 0.1 s by a player. The force on the hand is: 1

(a) 1000 N **(b)** 100 N **(c)** 50 N **(d)** 10 N

8. A block is pulled with a force of 15 N and moves with constant velocity. The friction on it is: 1

(a) 0 **(b)** 30 N **(c)** 7.5 N **(d)** 15 N

9. A man pushes a box of 20 kg with 50 N and the box moves at constant velocity. The friction is: 1

(a) 200 N **(b)** 50 N **(c)** 20 N **(d)** 0 N

10. A force acts on a mass of 2 kg for 4 s starting from rest and gives it a velocity of 8 m/s. The force is: 1

(a) 4 N **(b)** 2 N **(c)** 16 N **(d)** 8 N

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A body of mass 5 kg is acted on by two forces: 20 N east and 8 N west. Find the acceleration. 2

12. Find the force needed to stop a 2 kg body moving at 10 m/s in 4 s. 2

13. Why does a passenger in a car feel pushed to the right when the car turns left? 2

14. Find the weight of a 70 kg person on the Moon (g_moon = 1.7 m/s squared) and compare with that on Earth (g = 10). 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A body of mass 10 kg is acted on by a force of 50 N for 4 s starting from rest. Find the acceleration, the final velocity and the distance covered. 3

16. A truck of mass 5000 kg moving at 10 m/s collides and stops in 0.5 s. Find the average force and compare with its weight. 3

17. A hose pipe shoots water at 5 m/s at the rate of 2 kg/s on a wall. Find the force on the wall if the water stops. 3

OR

Explain, with a numerical example, how the time of impact affects the force: a 0.2 kg ball at 30 m/s is stopped in 0.02 s and in 0.2 s. 3

18. Explain why it is easier to push a box on rollers than on the bare floor, using the idea of friction. 3

OR

Explain why a ship takes a long time to stop. Use inertia and momentum. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A car of mass 1000 kg moves on a level road at 30 m/s. The engine force is 2000 N and the resistive force is 1500 N. Find the acceleration. If the engine is switched off and the resistance remains, find the retardation, the time to stop and the stopping distance. 5

OR

Explain the origin of friction, the factors on which it depends, and calculate the friction force on a 20 kg body with a coefficient of friction 0.3 (g = 10) and the acceleration if pulled with 80 N. 5

20. A gun of mass 4 kg fires a bullet of mass 20 g with a velocity of 400 m/s. Find the momentum of the bullet and the recoil velocity of the gun using equal and opposite momenta. 5

OR

A train of mass 200 tonnes moves at 72 km/h and is stopped by applying brakes over 400 m. Find the retardation, the time and the braking force. 5

END OF QUESTION PAPER — Total: 40 Marks